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NCFE · LEVEL 2 · MATHEMATICS

NCFE Functional Skills Mathematics Level 2

December 2019 Section B · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: NCFE-Maths-Level-2-Dec-2019-Section-B

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

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Activity 2: Running a business



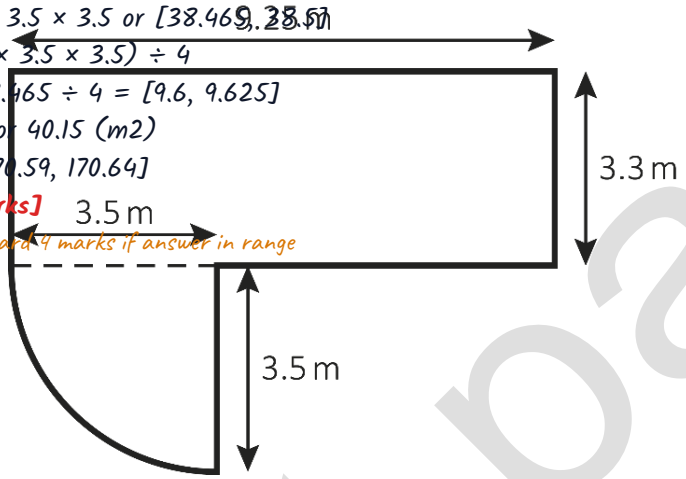
2 (a) Dan has set up a patio cleaning business.

He charges £4.25 per m^2 of the patio to be cleaned.

Mrs Jones has asked Dan how much he will charge for cleaning her patio.

Q2(a) Worked Solution:

This diagram shows her patio:



Not drawn accurately

(£)[170.60, 170.64]
3.14 × 3.5 × 3.5 or [38.465, 38.5]
(3.14 × 3.5 × 3.5) ÷ 4
OR 38.465 ÷ 4 = [9.6, 9.625]
40.14 or 40.15 (m2)
(£)[170.59, 170.64]
[4 marks]
Tip: Award 4 marks if answer in range

past paper

Examiner use only

How much will Dan charge Mrs Jones for cleaning her patio?

Use π = 3.14

[4 marks]

Your answer: **£**

past paper

Please turn over

Examiner use only

2 (b) Mr Patel pays Dan £250 to clean his patio.

The £250 includes VAT at 20%

How much was the bill before the VAT was added?

[2 marks]

Your answer: **£**

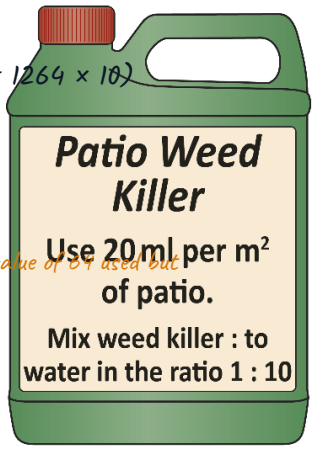
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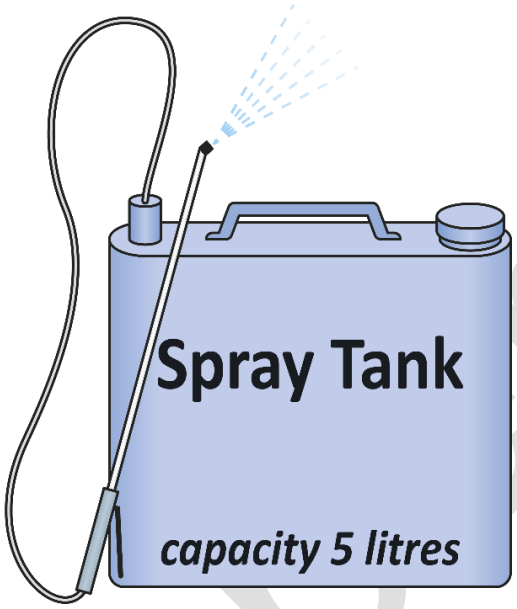
Examiner use only

2 (c) Dan needs to make a mixture of weed killer and water to clean Mr Patel's patio.

The area of the patio to be sprayed is 63.2 m^2

Q2(c) Worked Solution:
method 1 (working in ml)
6
20 × 63.2 or 1264
Their 1264 + (their 1264 × 10)
or 1264 + 12640
or 13904
2.78 or 2.8 or 3
[3 marks]
Tip: Award if rounded value of 64 used but





He fills the spray tank with the weed killer and water mixture.

How many times does Dan need to fill the spray tank to cover the whole patio?

[3 marks]

Your answer:

past paper

Please turn over

Examiner use only

2 (d) Dan wants to expand his business.

Next year he plans to clean patios from 8.30am to 3.30pm, Monday to Friday for 26 weeks during the spring and summer.

This table shows how long each job took Dan **this year**.
The times include travel time.

Time per cleaning job	Number of jobs
30 < time ≤ 60 minutes	25
60 < time ≤ 90 minutes	36
90 < time ≤ 120 minutes	45
120 < time ≤ 150 minutes	21
Total	127

Use the data to estimate how many patios Dan can clean **next year** if he works the hours planned.

[5 marks]

Your answer: **patios**

past paper

Examiner use only

2 (e) This table shows how long each job took Dan **last year**.

Time per cleaning job	Number of jobs
30 < time ≤ 60 minutes	14
60 < time ≤ 90 minutes	22
90 < time ≤ 120 minutes	30
120 < time ≤ 150 minutes	16
Total	82

Q2(e) Worked Solution:

Dan's accounts are checked by the tax office.

They pick one of **last year's** jobs at random to check.

or or 0.56 or 56%

82 41 What is the probability that the job would have taken more than 90 minutes?

(Calculator Test) **[1 mark]**

[1 mark]

Your answer:

[Total marks: 15]

past paper

Please turn over

Examiner use only

Activity 3: Cycling

3 (a) There are 51 million adults in the UK.

20.5 million do **not** do any regular exercise.

What fraction of adults in the UK do **not** do any regular exercise?

[1 mark]

Q3(a) Worked Solution:
41
102
[1 mark]
Tip: Do not accept

Your answer:

3 (b) The proportion of commuters who cycle to work in the UK is 0.06

Write 0.06 as:

- a fraction
- a percentage.

[2 marks]

Q3(b) Worked Solution:
6 3
or
100 50
6(%)
[2 marks]

Your answer: **Fraction:**

Percentage:

past paper

Please turn over

Examiner use only

3 (c) Jackie is training for a cycle race.
She reads a cycling blog.

It says that cyclists should aim to keep their heart rate above 72% of their maximum heart rate.

Jackie uses this formula to work out her maximum heart rate:

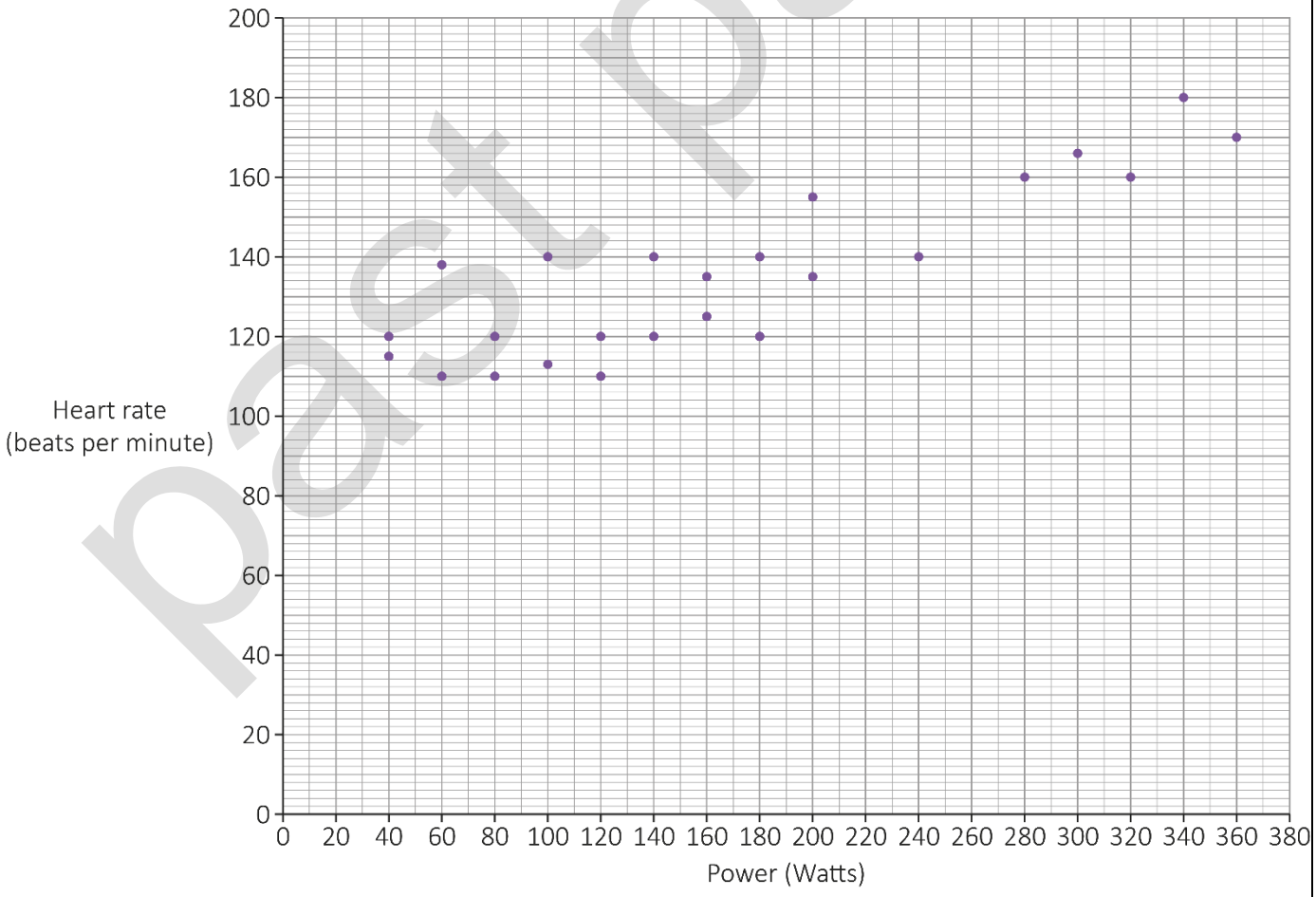
$$M = 191.5 - 0.007A^2$$

Where: M is the maximum heart rate in beats per minute
A is age in years.

Jackie is 25 years old.

This scatter diagram shows the relationship between power (in Watts) and heart rate (beats per minute):

Relationship between power (in Watts) and heart rate (in beats per minute)



past paper

Examiner use only

To produce 100 Watts of power, a cyclist uses about 350 kilocalories per hour.

Estimate the number of kilocalories Jackie uses per hour if she cycles at 72% of her maximum heart rate.

[6 marks]

Your answer: **kilocalories**

past paper

Please turn over

Examiner use only

3 (d) Rafa and his friends cycle to keep fit.

They want to reduce the mean time it takes them to cycle 10 km

Their target is to reduce the time by 11% within 12 weeks.

This table shows their times in **Week 1** and **Week 12**.

	Time (minutes) to cycle 10 km	
	Week 1	**Week 12**
Ann	26.3	23.9
Mike	22.0	20.2
Rafa	28.1	25.7
Ali	24.9	20.4
Maria	18.4	18.1
Luke	21.7	20.7
Becky	20.3	17.3

Has the group achieved its target?

Show how you decide.

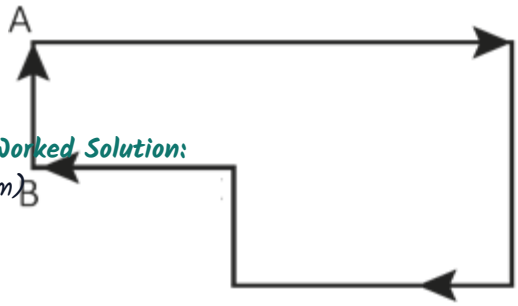
[4 marks]

Your answer:

past paper

Examiner use only

3 (e) Peta cycles around this route:



Not drawn accurately

Q3(e) Worked Solution:
20 (km)
20
23
9

She starts at A.

When she reaches B she has cycled 12.5 miles.

(i) What is 12.5 miles in km?

Use this conversion: 1.6 km = 1 mile

[1 mark]

(Calculator Test)
[2 marks]

Your answer:

The total route is 23 km

(ii) What fraction of the total route has she cycled when she reaches B?

[1 mark]

Your answer:

[Total marks: 15]

Please turn over

past paper

Examiner use only

Activity 4: Packaging

4 (a) Beth is on a work placement.
The company makes biscuits.

In 2018 it reduced the number of biscuits in each packet.

The weight of each biscuit stayed the same.

Q4(a) Worked Solution:

The weight of a packet went from 135 g to 108 g

5 (biscuits)

The 135 g packet contained 25 biscuits.

How many fewer biscuits did the 108 g packet contain?

[2 marks]

(108 ÷ 135) × 25
or 108 ÷ (135 ÷ 25)
or 20
[2 marks]
Tip: Award 2 marks if correct answer given

Your answer: **biscuits**

past paper

Examiner use only

4 (b) The company also makes pizza.

This table shows information about the two sizes it sells:

Pizza	Diameter	Depth
Large	38 cm	2 cm
Small	19 cm	1.5 cm

Q4(b) Worked Solution:

Beth says that the large size gives over 5 times as much pizza as the small size.

Is Beth correct?

Show how you decide.

Use $\pi = 3.14$

[4 marks]

5.3(...) or 2125(...) AND yes
3.14 × 19 × 19 × 2 or 2267(.08)
or 3.14 × 9.5 × 9.5 × 1.5 or 425(.0775)
2267(.08) and 425(.0775)
Their 2267(.08) ÷ their 425(.0775)
or 5 × their 425(.0775)
or 5(.3) or 2125(...)
[4 marks]
Tip: Award 4 marks if correct answer given

Your answer:

past paper

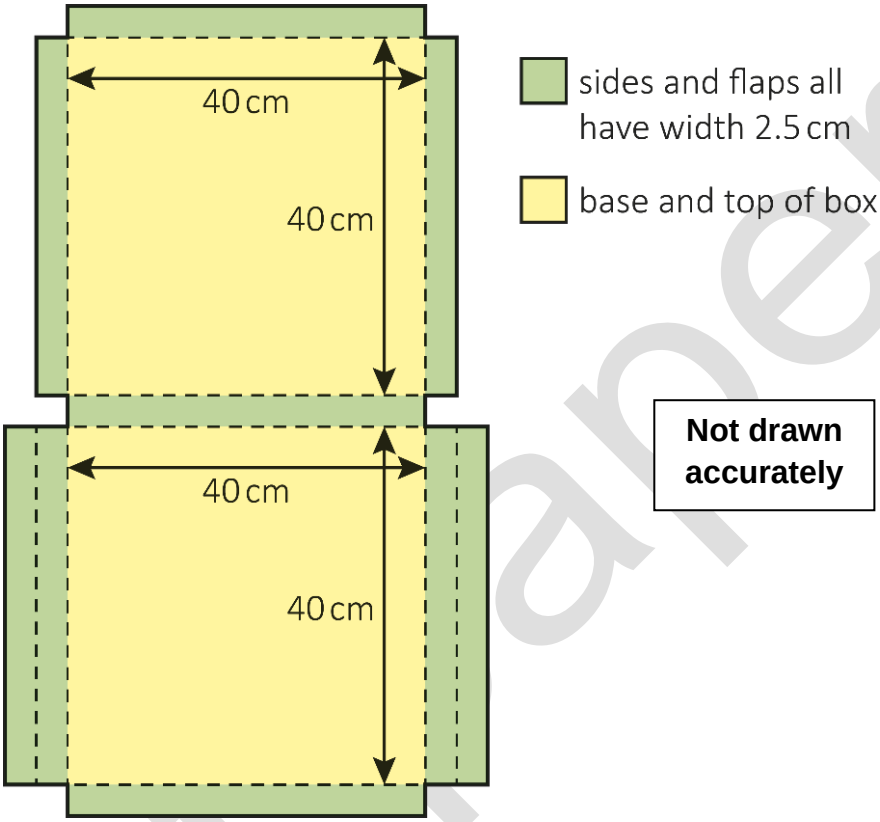
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Examiner use only

4 (c) The large pizzas are sold in cardboard boxes.

When closed, the boxes measure 40 cm by 40 cm by 2.5 cm and are cuboid.

This is the net of the pizza box:



past paper

Examiner use only

The company thinks it could use the same net but reduce the dimensions to 38 cm by 38 cm by 2 cm

Beth is asked to calculate the percentage reduction in cardboard if the smaller boxes are used.

What answer should Beth get?

[5 marks]

Your answer: **%**

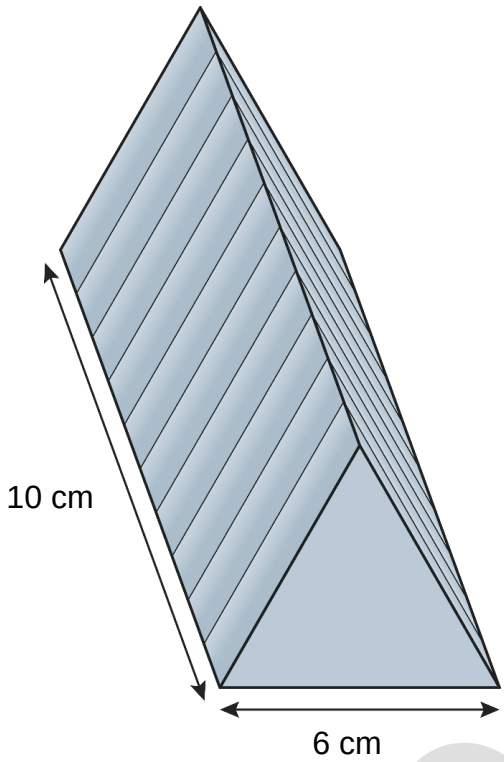
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Examiner use only

4 (d) The company also makes a chocolate bar.

It is divided into 10 identical triangle-shaped pieces.



Not drawn accurately

Draw a plan view of the bar in the space below.

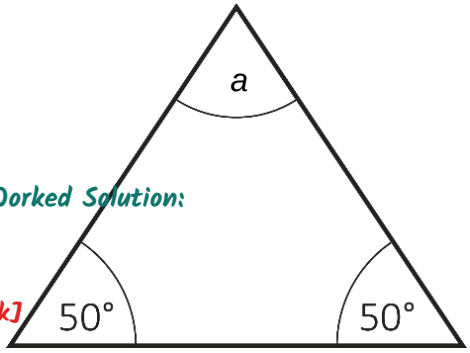
Label the dimensions.

[3 marks]

past paper

Examiner use only

4 (e) One triangular face of the chocolate bar is shown in the diagram.



Q4(e) Worked Solution:
80 (°)
//
[1 mark]

Not drawn accurately

Calculate the size of the angle labelled *a*

[1 mark]

Your answer: °

[Total marks: 15]

This is the end of the assessment.

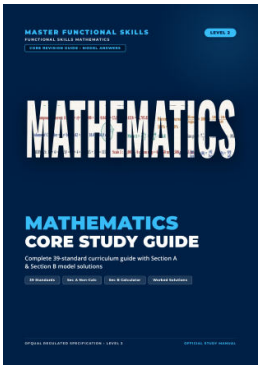
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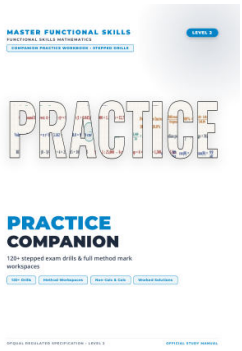


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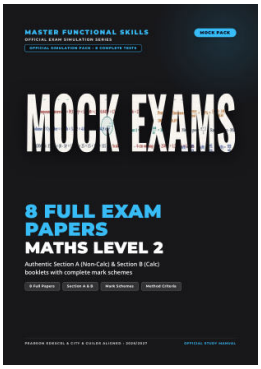


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